VEHICLE SERVICE MANAGEMENT SYSTEM

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PROJECT OVERVIEW

Automobile service centers rely on manual or semi-automated processes, making it difficult to track service history, schedule technicians efficiently, manage spare parts, and monitor workload and revenue. This lack of a centralized system results in delays, poor visibility, and reduced customer satisfaction, creating the need for a secure and scalable Vehicle Service Management System.

Solution :

- Centralized platform to manage vehicle service requests and complete service history.
- Automated technician and service bay assignment based on workload and specialization.
- Real-time inventory and spare parts tracking with low-stock alerts.
- End-to-end service lifecycle tracking from booking to closure.
- Role-based dashboards for Admin, Manager, Technician, and Customer.
- Automated billing, invoicing, and email notifications for better customer experience.

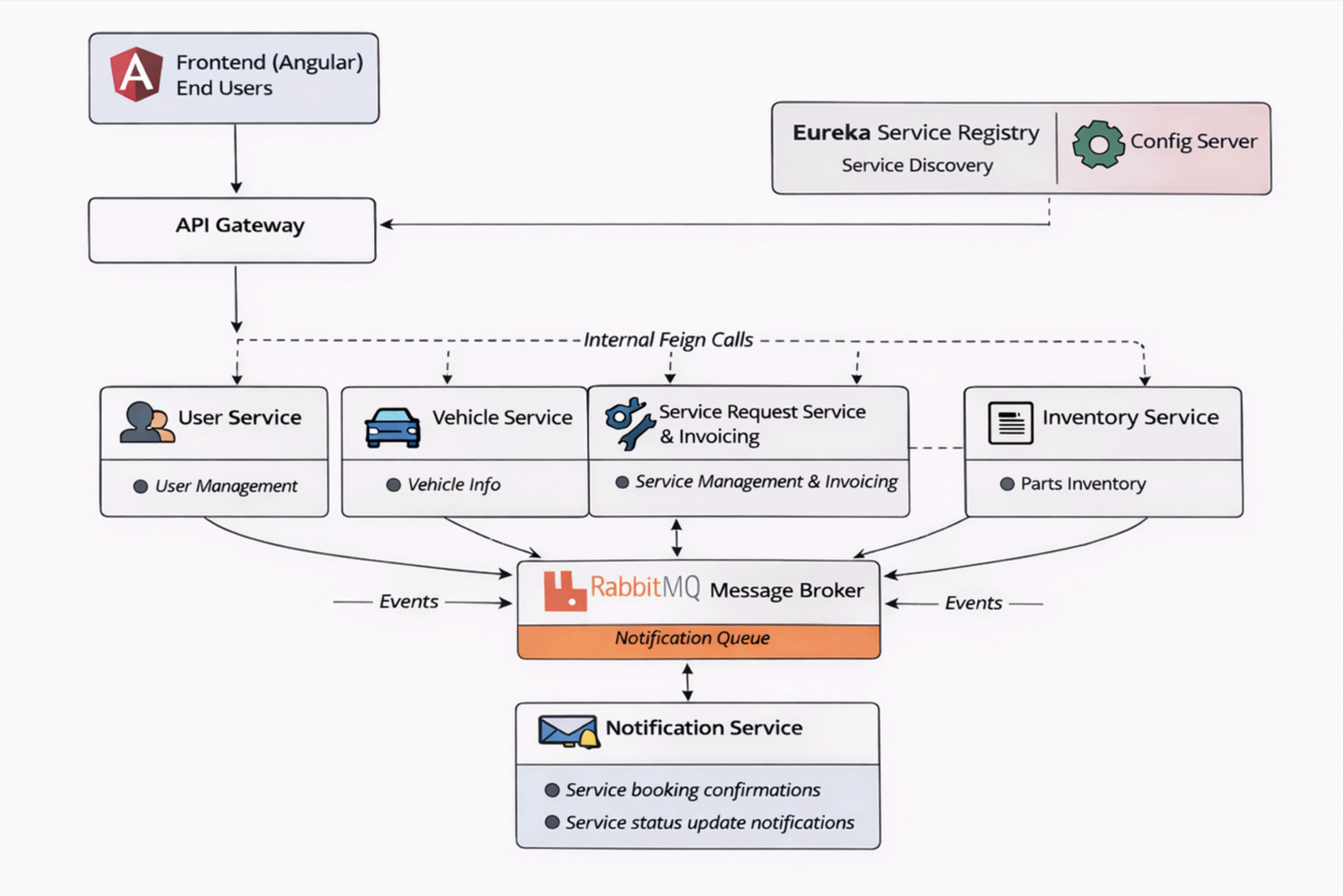
MICROSERVICES ARCHITECTURE

8 Microservices:

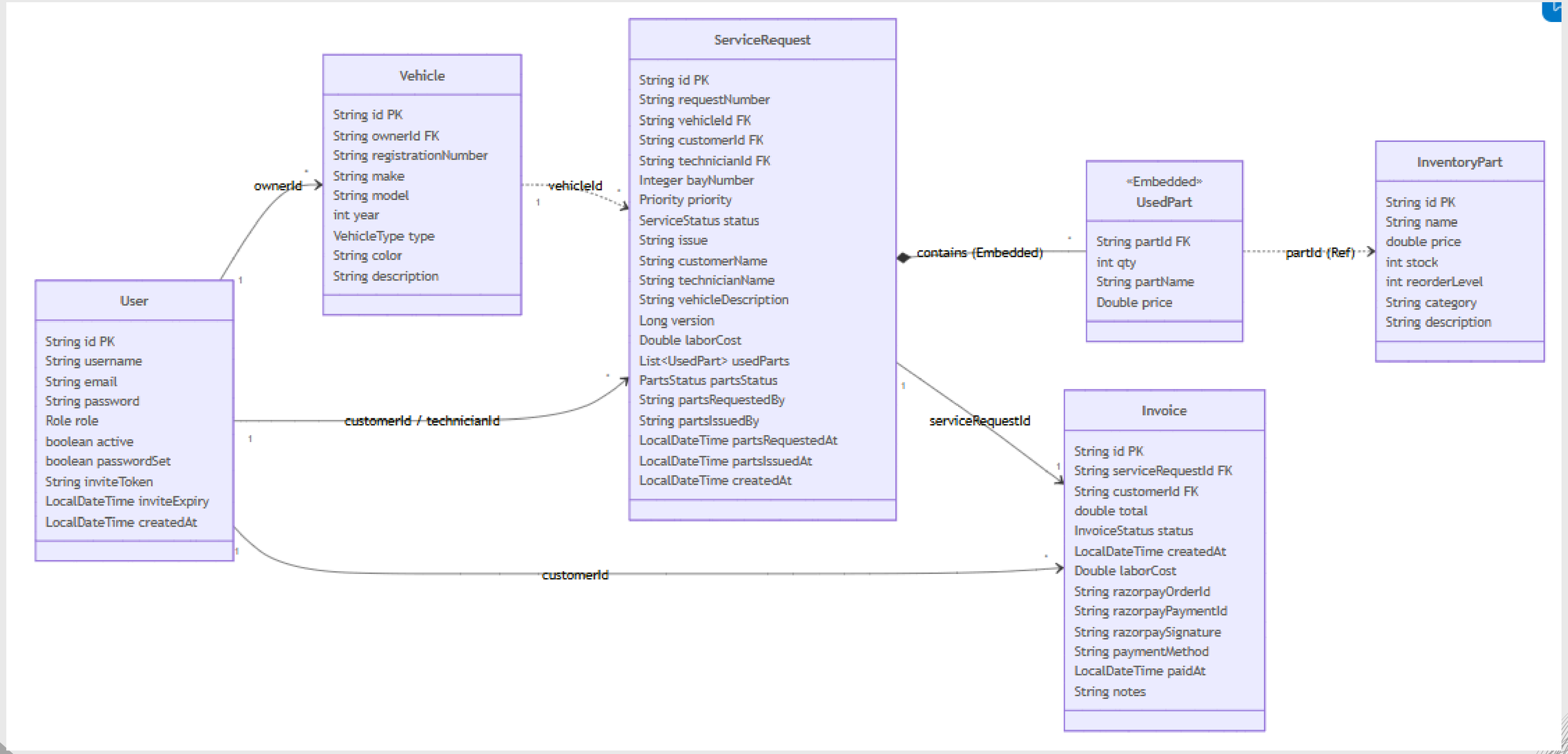
1.Config Server (8888) - Centralized configuration
2.Eureka Server (8761) - Service discovery
3.API Gateway (8080) - Entry point, JWT authentication, routing
4.User Service - Authentication, user management
5.Vehicle Service - Vehicle registration
6.Inventory Service - Spare parts, part requests
7.Service Request Service - Service lifecycle, billing, bays
8.Notification Service - Email notifications
Inter-Service Communication:

- Feign Clients
- RabbitMQ

MICROSERVICES ARCHITECTURE



DATABASE SCHEMA



USER ROLES & RESPONSIBILITIES

1. **Admin :**
- Approves managers and technicians, manages user accounts.
- Controls system settings, pricing, and overall monitoring.
2. **Manager :**
- Assigns service requests to technicians and service bays.
- Tracks workload, availability, and service progress.
3. **Technician :**
- Performs assigned services and updates job status.
- Adds remarks and marks services as completed.
4. **Customer :**
- Books vehicle services and tracks service status.
- Views service history and invoices.

SERVICE REQUEST WORKFLOW

1.Book Service (Customer)
- Creates request (eg : SR-1234) → Status: **REQUESTED**

2.Assign Job (Manager)
- Selects Technician & Bay → Status: **ASSIGNED**

3.Execution & Parts (Technician)
- Starts Job → Status: **IN_PROGRESS**
- Requests Parts → Manager Approves → *PARTS_APPROVED*

4.Completion (Technician)
- Finishes work → Status: **COMPLETED**

5.Close & Bill (Manager)
- Finalizes labor cost.
- System Actions: Deducts Inventory Stock + Generates Invoice.
- Status: **CLOSED**

6.Payment (Customer)
- Pays via Razorpay → Invoice Status: *PAID*

KEY FEATURES IMPLEMENTED

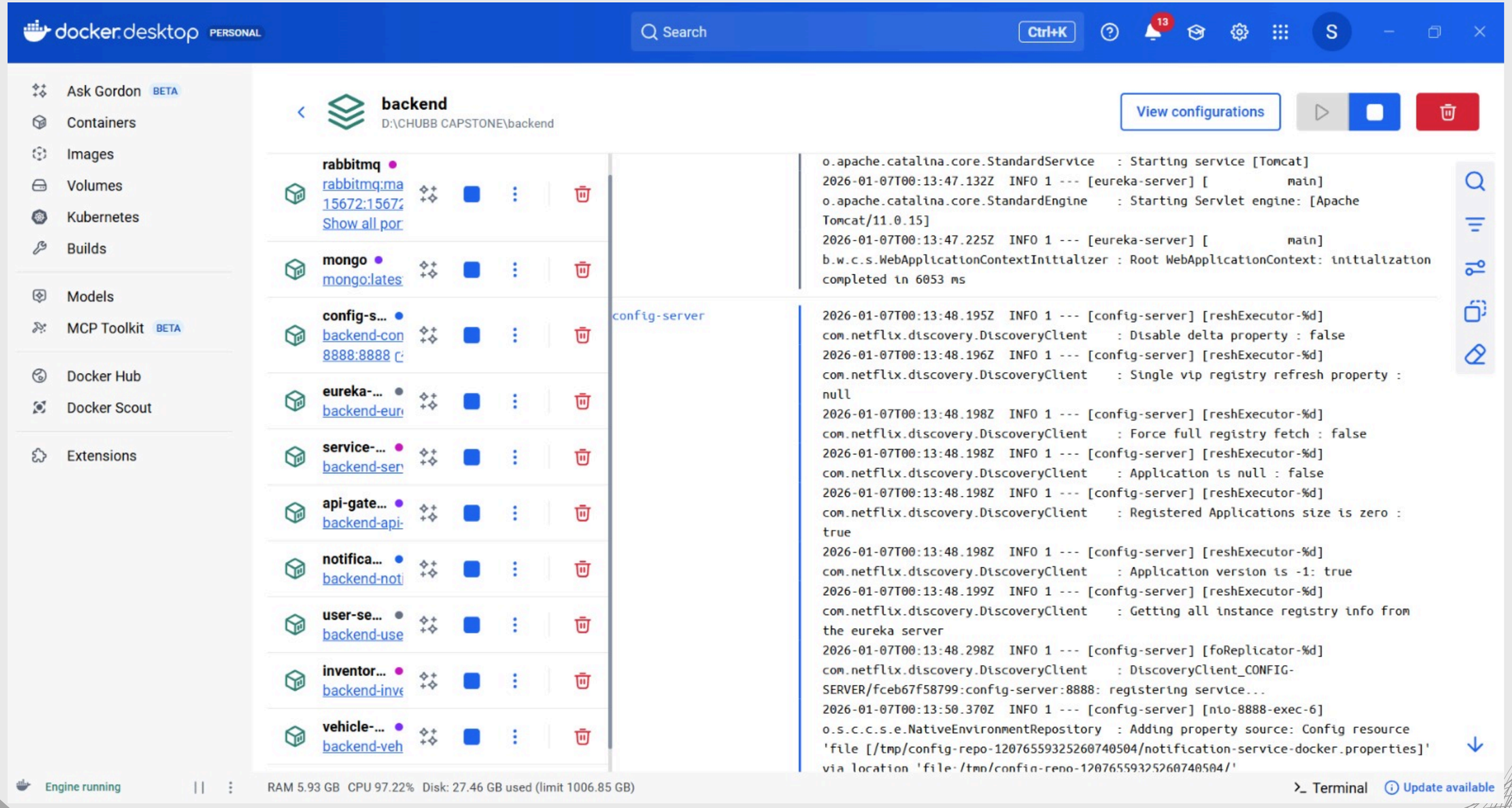
Feature Module	Functionality	How It Works (Implementation)
Onboarding	**Secure Account Activation**	Admins create accounts which stay **Inactive** until the user clicks a unique, time-sensitive **Invitation Link** (Token-based) to set their password.
Smart Assignment	**Workload Balancing**	System **automatically blocks** assignment if a Technician has reached their daily job capacity (e.g., 5 jobs/day).
Resource Management	**Bay Conflict Resolution**	Real-time validation prevents assigning multiple vehicles to the same **Service Bay** simultaneously.
Inventory Control	**2-Step Parts Verification**	Technicians must **Request** parts, but stock is only deducted after a Manager **Approves** them, preventing theft/waste.
Payments	**Razorpay Integration**	Secure payment gateway with **Signature Verification** to ensure transaction authenticity before updating invoice status.
Analytics	**Technician Statistics**	Real-time dashboards showing **Active Workload** (Current Jobs) vs. **Performance History** (Completed Jobs) for Manager oversight.
Resilience	**Zero-Downtime Alerts**	**Assigned** jobs proceed even if Inventory Service is down (Circuit Breaker), syncing stock data later.

CI/CD PIPELINE

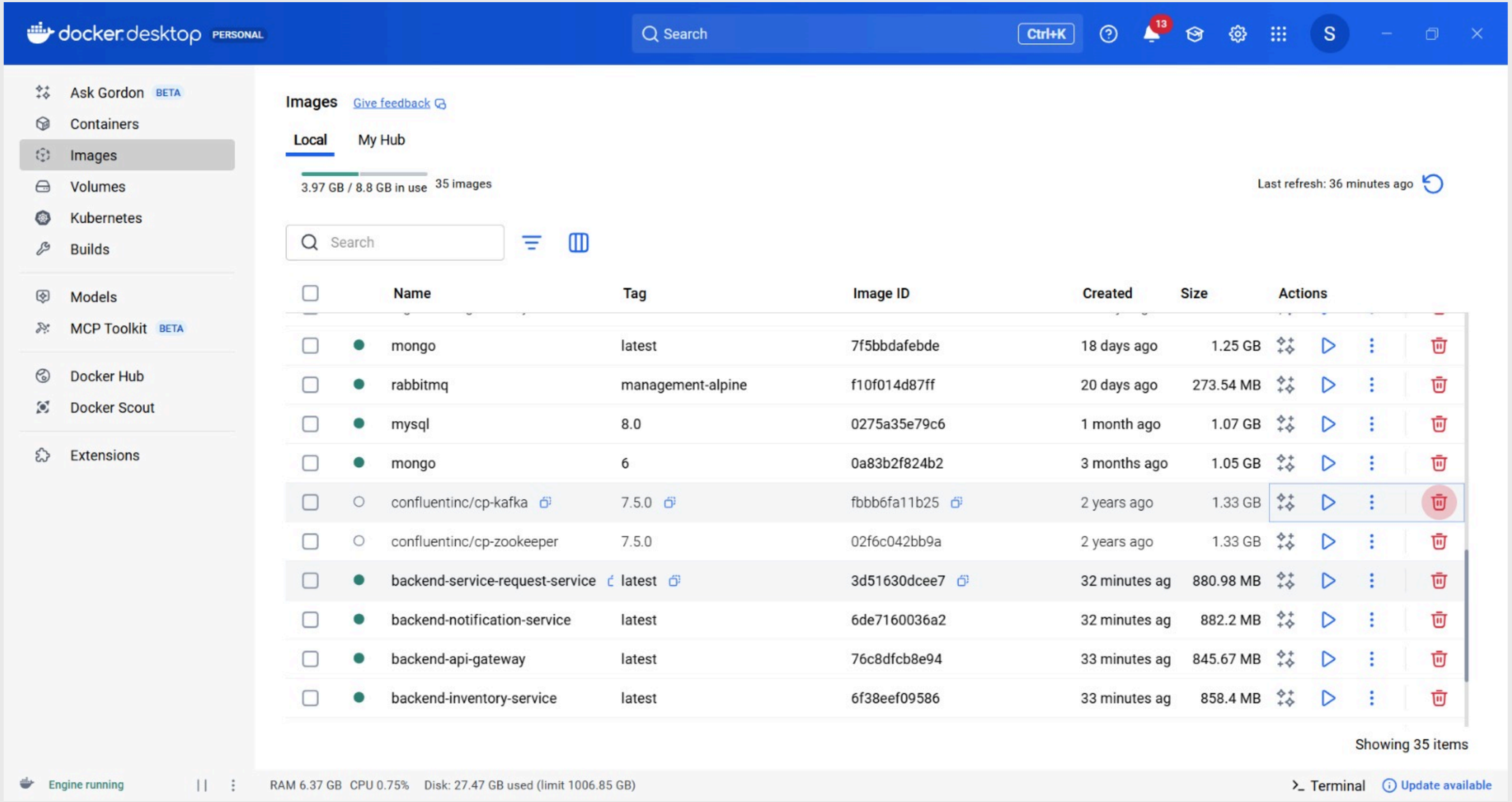
Jenkins / vehicle-service-managem... / #7 / Console Output

```
Container user-service  Starting
Container api-gateway  Starting
Container notification-service  Starting
Container service-request-service  Starting
Container inventory-service  Started
Container vehicle-service  Started
Container user-service  Started
Container api-gateway  Started
Container service-request-service  Started
Container notification-service  Started
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Pipeline completed successfully!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

DOCKER


docker desktop PERSONAL
Search
Ctrl+K
Ask Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit BETA
Docker Hub
Docker Scout
Extensions
backend
D:\CHUBB CAPSTONE\backend
View configurations
rabbitmq
mongo
config-s...
eureka-...
service-...
api-gate...
notifica...
user-se...
inventor...
vehicle-...
config-server
o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-01-07T00:13:47.132Z INFO 1 --- [eureka-server] [main]
o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.15]
2026-01-07T00:13:47.225Z INFO 1 --- [eureka-server] [main]
b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 6053 ms
2026-01-07T00:13:48.195Z INFO 1 --- [config-server] [reshExecutor-%d]
com.netflix.discovery.DiscoveryClient : Disable delta property : false
2026-01-07T00:13:48.196Z INFO 1 --- [config-server] [reshExecutor-%d]
com.netflix.discovery.DiscoveryClient : Single vip registry refresh property : null
2026-01-07T00:13:48.198Z INFO 1 --- [config-server] [reshExecutor-%d]
com.netflix.discovery.DiscoveryClient : Force full registry fetch : false
2026-01-07T00:13:48.198Z INFO 1 --- [config-server] [reshExecutor-%d]
com.netflix.discovery.DiscoveryClient : Application is null : false
2026-01-07T00:13:48.198Z INFO 1 --- [config-server] [reshExecutor-%d]
com.netflix.discovery.DiscoveryClient : Registered Applications size is zero : true
2026-01-07T00:13:48.198Z INFO 1 --- [config-server] [reshExecutor-%d]
com.netflix.discovery.DiscoveryClient : Application version is -1: true
2026-01-07T00:13:48.199Z INFO 1 --- [config-server] [reshExecutor-%d]
com.netflix.discovery.DiscoveryClient : Getting all instance registry info from the eureka server
2026-01-07T00:13:48.298Z INFO 1 --- [config-server] [foReplicator-%d]
com.netflix.discovery.DiscoveryClient : DiscoveryClient_CONFIG-SERVER/fceb67f58799:config-server:8888: registering service...
2026-01-07T00:13:50.370Z INFO 1 --- [config-server] [nio-8888-exec-6]
o.s.c.c.s.e.NativeEnvironmentRepository : Adding property source: Config resource 'file [/tmp/config-repo-12076559325260740504/notification-service-docker.properties]'
Engine running
RAM 5.93 GB CPU 97.22% Disk: 27.46 GB used (limit 1006.85 GB)
Terminal
Update available

DOCKER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Search
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13
S
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Images
Volumes
Kubernetes
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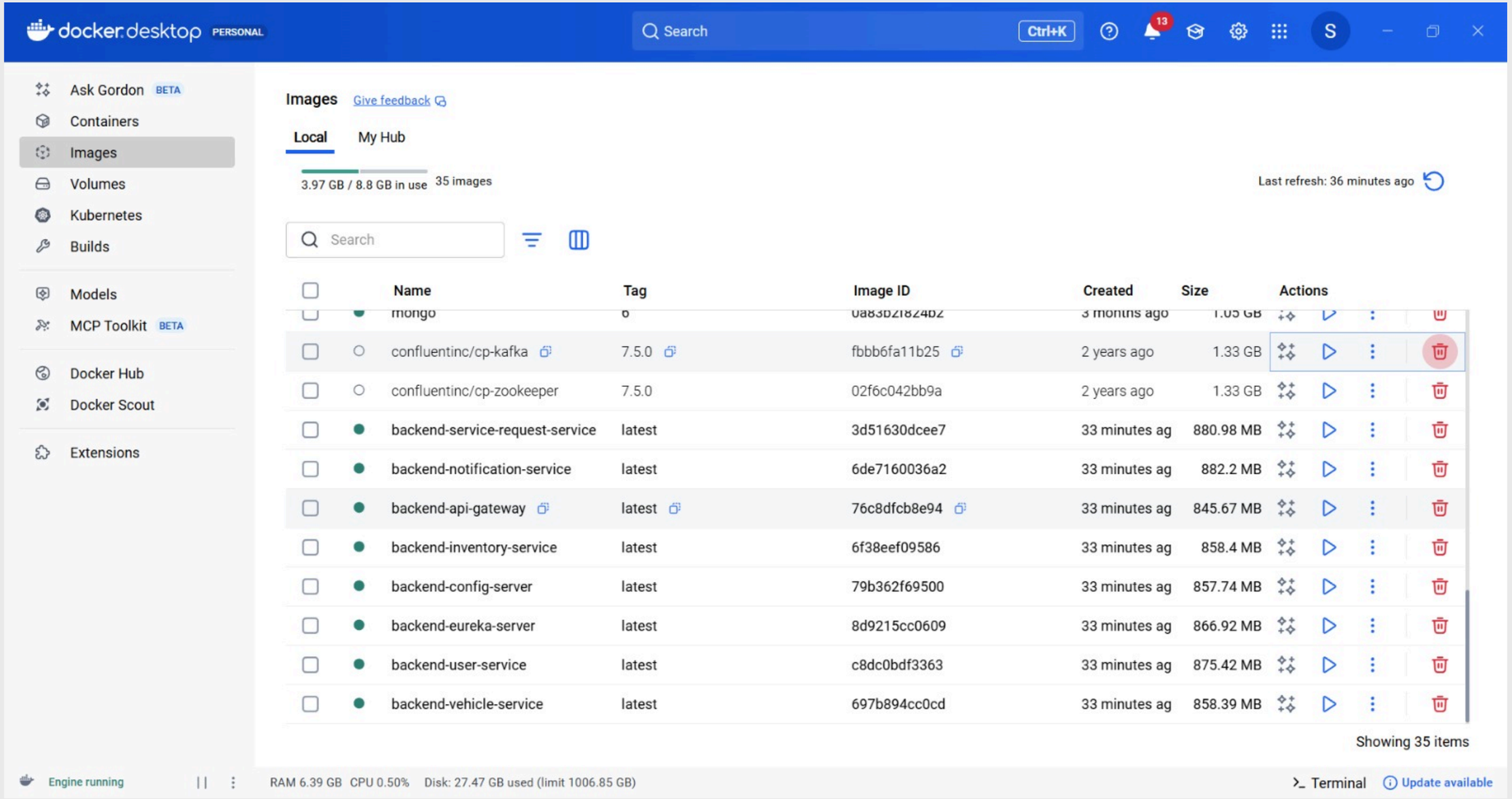
Models
MCP Toolkit BETA
Docker Hub
Docker Scout
Extensions
Images
Give feedback
Local
My Hub
3.97 GB / 8.8 GB in use
35 images
Last refresh: 36 minutes ago
Search
Name
Tag
Image ID
Created
Size
Actions
mongo latest 7f5bbdafebde 18 days ago 1.25 GB
rabbitmq management-alpine f10f014d87ff 20 days ago 273.54 MB
mysql 8.0 0275a35e79c6 1 month ago 1.07 GB
mongo 6 0a83b2f824b2 3 months ago 1.05 GB
confluentinc/cp-kafka 7.5.0 fbbb6fa11b25 2 years ago 1.33 GB
confluentinc/cp-zookeeper 7.5.0 02f6c042bb9a 2 years ago 1.33 GB
backend-service-request-service latest 3d51630dcee7 32 minutes ag 880.98 MB
backend-notification-service latest 6de7160036a2 32 minutes ag 882.2 MB
backend-api-gateway latest 76c8dfcb8e94 33 minutes ag 845.67 MB
backend-inventory-service latest 6f38eef09586 33 minutes ag 858.4 MB
Showing 35 items
Engine running
RAM 6.37 GB CPU 0.75% Disk: 27.47 GB used (limit 1006.85 GB)
Terminal
Update available

DOCKER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My Hub
3.97 GB / 8.8 GB in use
35 images
Last refresh: 36 minutes ago
Search
Name
Tag
Image ID
Created
Size
Actions
confluentinc/cp-kafka 7.5.0 fbbb6fa11b25 2 years ago 1.33 GB
confluentinc/cp-zookeeper 7.5.0 02f6c042bb9a 2 years ago 1.33 GB
backend-service-request-service latest 3d51630dcee7 33 minutes ag 880.98 MB
backend-notification-service latest 6de7160036a2 33 minutes ag 882.2 MB
backend-api-gateway latest 76c8dfcb8e94 33 minutes ag 845.67 MB
backend-inventory-service latest 6f38eef09586 33 minutes ag 858.4 MB
backend-config-server latest 79b362f69500 33 minutes ag 857.74 MB
backend-eureka-server latest 8d9215cc0609 33 minutes ag 866.92 MB
backend-user-service latest c8dc0bdf3363 33 minutes ag 875.42 MB
backend-vehicle-service latest 697b894cc0cd 33 minutes ag 858.39 MB
Showing 35 items
Engine running
RAM 6.39 GB CPU 0.50% Disk: 27.47 GB used (limit 1006.85 GB)
Terminal
Update available

SUMMARY

- Microservices Architecture with 8 independent services
- Secure JWT-based Authentication & Role-Based Access Control .
- Secure Account Activation (Token-based) & Razorpay Payment Integration.
- Complete Service Lifecycle Management (Pending → Assigned → In Progress → Closed).
- Smart Workload Balancing & Real-time Bay Conflict Resolution.
- Spare Parts Inventory Management with 2-Step Verification (Request → Approve).
- Real-time Email Notifications via RabbitMQ
- Automated Billing & Dynamic Invoice Generation.
- CI/CD Pipeline with Jenkins, Docker & SonarCloud.
- 90%+ Code Coverage with JUnit & Mockito.

THANK YOU